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Coding Patterns

Week 5 — Depth over breadth: patterns as decision tools

Strategy | Observer | Adapter | Repository | Command | State

What you'll learn

- Why patterns exist (recurring pain → reusable structure)
- How to choose a pattern using practical "selection lenses"
- How to adopt patterns incrementally without a rewrite

Depth over breadth

Patterns reduce risk — mastering the “why” builds reusable judgment.

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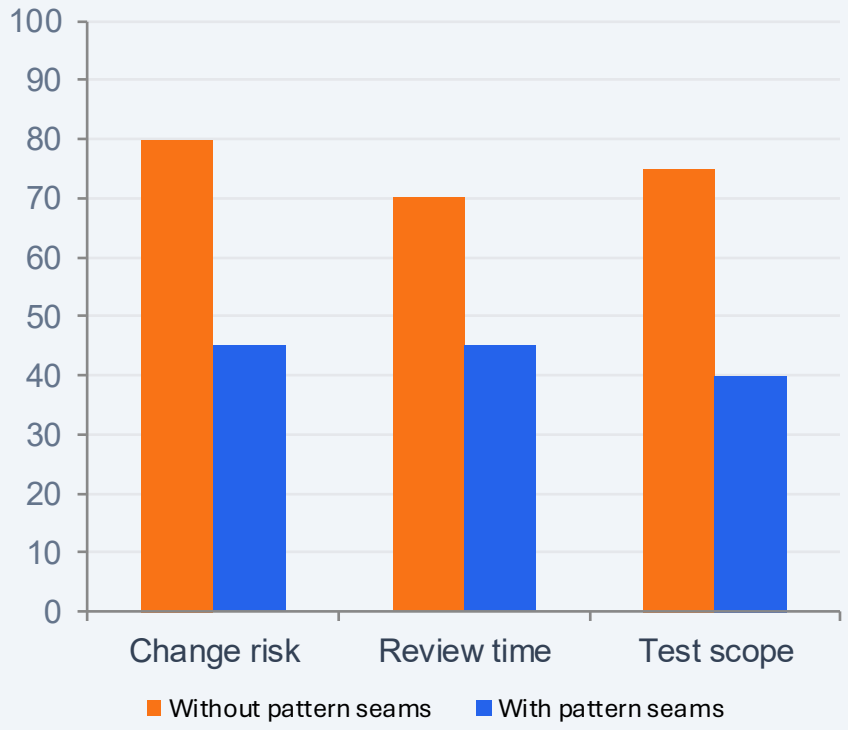
Why fewer patterns?

- Patterns are tools: pick the smallest one that removes today’s pain.
- Consistency speeds onboarding, reviews, and refactors.
- Real examples reveal where teams lose time and reliability.

Illustrative impact (not real data)



Goal: safe change speed

Agenda

A practical walk-through: problem → pattern → tradeoffs → adoption.

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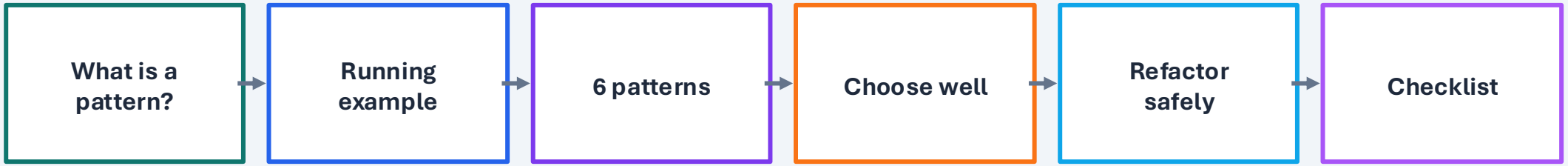
Lesson flow



- We'll compare similar patterns so you can choose based on requirements.
- You'll see incremental refactoring paths (no rewrites).

Anatomy of a pattern

Patterns aren't templates — they are tradeoff-aware solution structures.

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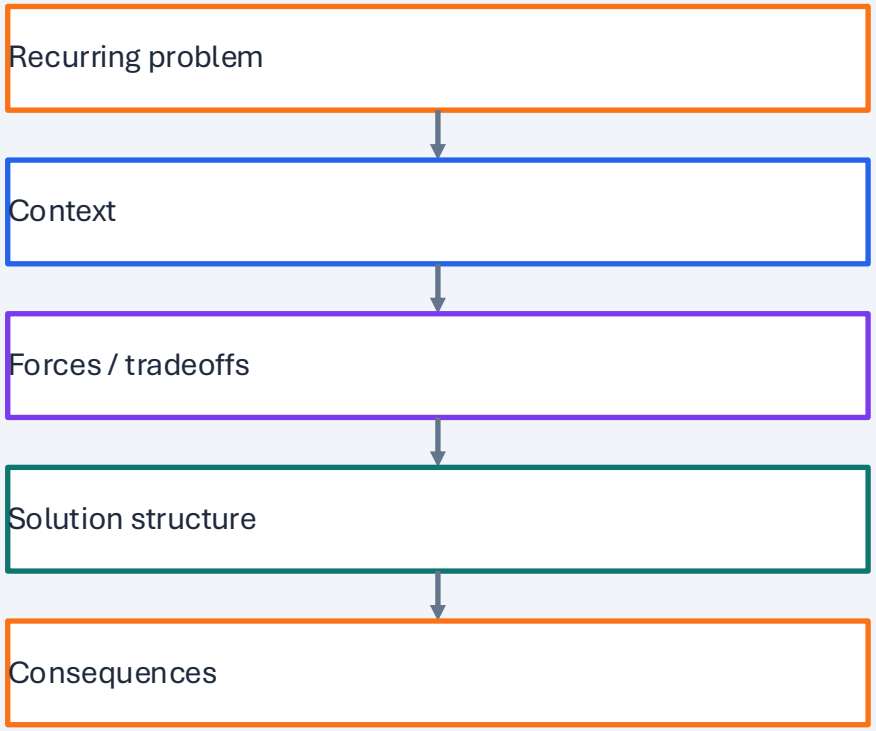
A pattern includes...

- Problem that keeps reappearing
- Context where it's safe to apply
- Forces & tradeoffs (why it's not one-size-fits-all)
- Solution structure you can adapt
- Consequences (the cost you're paying)

Cargo-cult prevention



If you can't name the problem + tradeoff, you're copying, not designing.

Why patterns exist

Recurring pain → predictable structure.

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Example pain: enrollment rules grow every semester

- Rules pile into long if/else chains across files.
- One tweak breaks an unrelated rule (hidden coupling).
- Tests become brittle when logic mixes with database calls.
- Releases slow down because behavior is hard to explain.

From chaos → seams

Before

EnrollmentService
+ if/else rules
+ SQL calls

Hard to test

After

Pricing (Strategy)

↓

Data access (Repository)

↓

Notifications (Observer)

Small units → fast tests

Pattern selection lenses

Use these questions to justify a pattern (instead of copying one).

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Six lenses

Change frequency

What changes often (rules, vendors, states)?

Ownership boundaries

Who owns each part? How many teams coordinate?

Testability

Can we test logic without DB / network?

Runtime constraints

Is indirection acceptable in hot paths?

Failure modes

Can one failure block the whole action?

Observability

Can we trace behavior with logs + IDs?

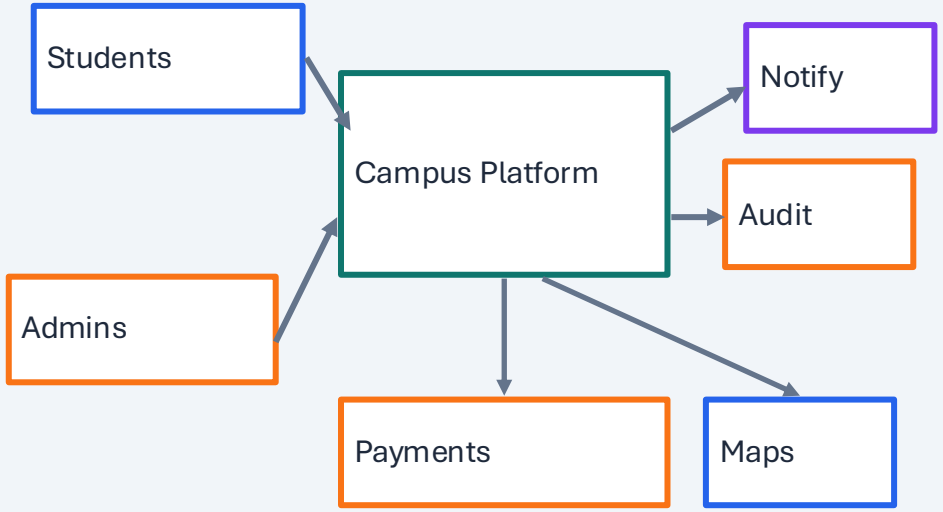
Running example: Campus Services Platform

Enrollment, shuttles, and events — with real integrations & audit needs.

Scope (student-sized, real-world pain)

- Different student categories → different pricing & eligibility rules.
- Notifications when seats open or routes change.
- Third-party payments and map providers.
- Admins must audit and undo enrollment changes.

System context



This single example generates all six pattern problems.

Strategy pattern — problem

Strategy

Pricing and eligibility rules multiply over time.

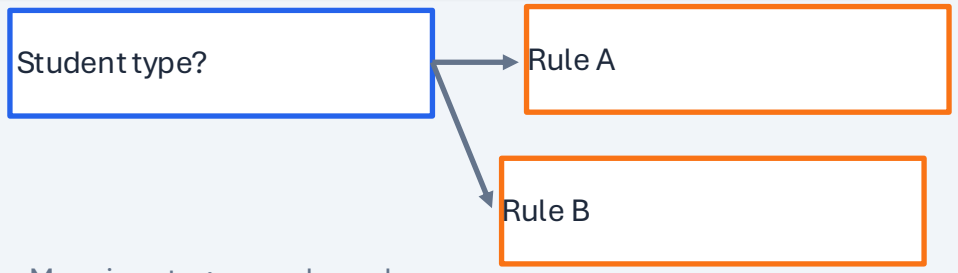
Symptoms

- One pricing function grows into nested if/else.
- Adding a scholarship means editing core logic in many places.
- Bugs appear because rule order matters.
- Testing is slow: you must exercise every branch combo.

Branch explosion

Typical outcome

```
function price(ctx) {
  if (ctx.student === 'staff') {
    if (ctx.event === 'sports') return ...
    else if (ctx.event === 'concert') return ...
  } else if (ctx.student === 'scholarship') {
    if (ctx.window === 'early') return ...
    else return ...
  }
  // ...more rules...
}
```



More inputs → more branches

Strategy pattern — solution

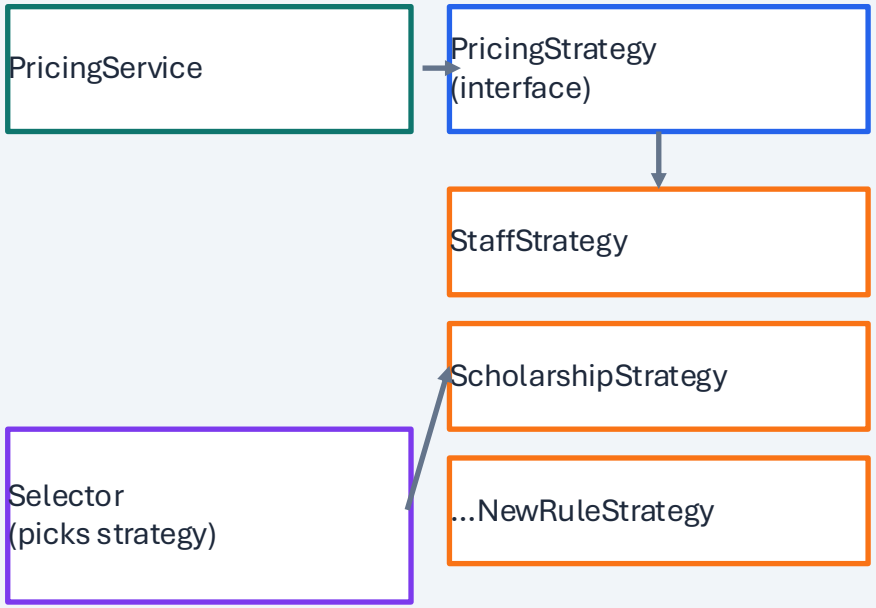
Strategy

Encapsulate each rule behind a shared interface.

How it helps

- Each rule becomes a strategy with a shared interface.
- Selection happens at runtime based on context.
- New rules = add a class, not edit the core.
- Strategies unit-test in isolation (lower regression risk).

Structure



Cost: more files. Benefit: safe change speed.

Strategy pattern — code example

Strategy

Each pricing rule is a separate class behind a shared interface.

```
// 1. Define the strategy interface
interface PricingStrategy {
  calculate(ctx: EnrollmentContext): Price;
}

// 2. Concrete strategies — one per rule
class StaffPricing implements PricingStrategy {
  calculate(ctx) { return ctx.basePrice * 0.5;  }  // 50% staff discount
}
class ScholarshipPricing implements PricingStrategy {
  calculate(ctx) { return 0;  }  // fully covered
}

// 3. Selector picks the right strategy at runtime
const strategies: Record<string, PricingStrategy> = {
  staff:       new StaffPricing(),
  scholarship: new ScholarshipPricing(),
};

// 4. Service uses the strategy — no if/else needed
function price(ctx: EnrollmentContext): Price {
  const strategy = strategies[ctx.studentType];
  return strategy.calculate(ctx);  // no branches!
}
```

Observer pattern — problem

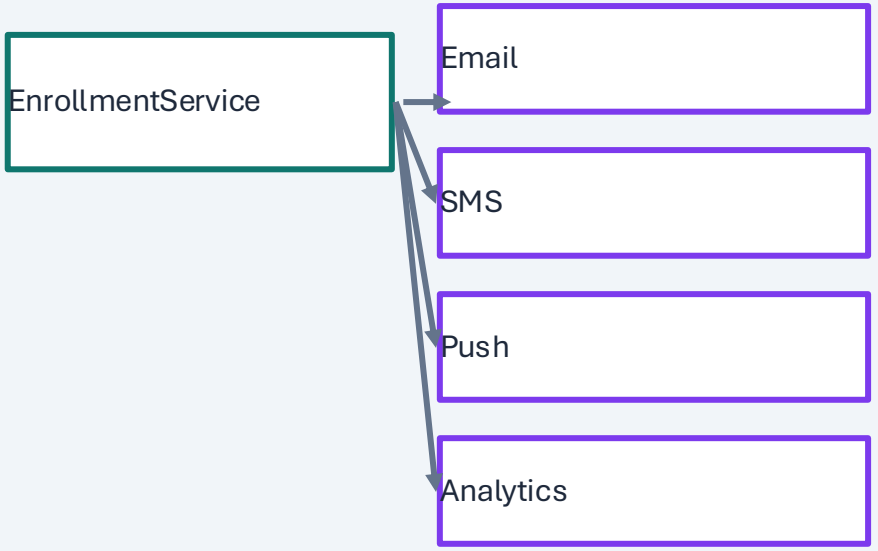
Observer

One event triggers many side effects — coupling grows fast.

Symptoms

- Enrollment service directly calls Email, SMS, Push modules.
- New channel → core service must change.
- One channel failure can block the entire action.
- Teams avoid adding features because integration points are fragile.

Tight coupling



Every arrow is a change + failure risk surface.

Observer pattern — solution

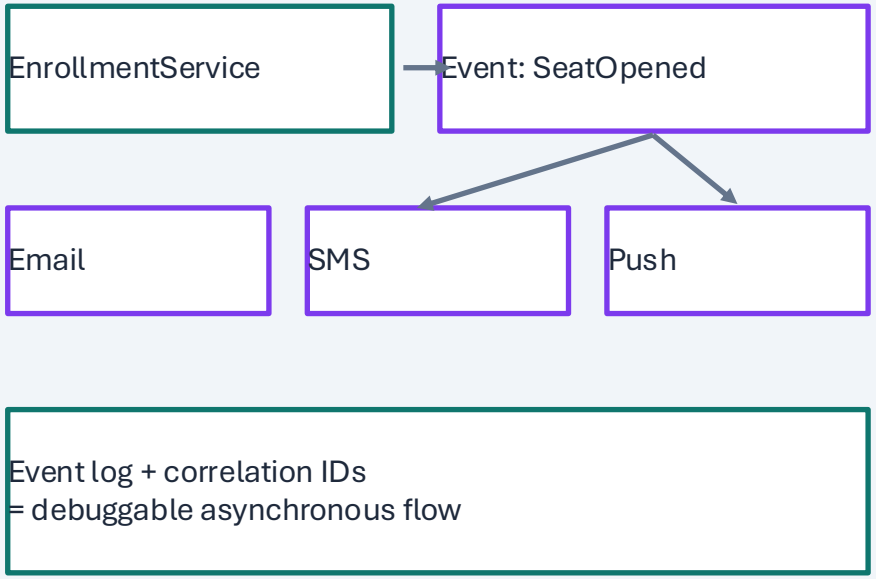
Observer

Publish an event; subscribers react independently.

How it helps

- Enrollment emits SeatOpened and moves on.
- Channels subscribe — add new ones without editing the core.
- Failures stay local; one bug won't stop others.
- Event logs become an audit-friendly trace.

Event-driven structure



Observer pattern — code example

Observer

Emit events; subscribers react independently without coupling.

```
// 1. Define an event type and a subscriber interface
type SeatEvent = { courseId: string; studentId: string };
interface SeatObserver { onSeatOpened(e: SeatEvent): void; }

// 2. Concrete subscribers — each handles one concern
class EmailNotifier implements SeatObserver {
  onSeatOpened(e) { sendEmail(e.studentId, "Seat available"); }
}
class AnalyticsTracker implements SeatObserver {
  onSeatOpened(e) { trackEvent("seat_opened", e); }
}

// 3. The service publishes events — knows nothing about subscribers
class EnrollmentService {
  private observers: SeatObserver[] = [];

  subscribe(obs: SeatObserver) { this.observers.push(obs); }

  dropStudent(courseId: string, studentId: string) {
    // ... remove student from course ...
    const event = { courseId, studentId };
    for (const obs of this.observers) obs.onSeatOpened(event);
  }
}

// 4. Wire it up — add new channels without touching the service
const svc = new EnrollmentService();
svc.subscribe(new EmailNotifier());
svc.subscribe(new AnalyticsTracker()); // add more without changing the service
```

Adapter pattern — problem

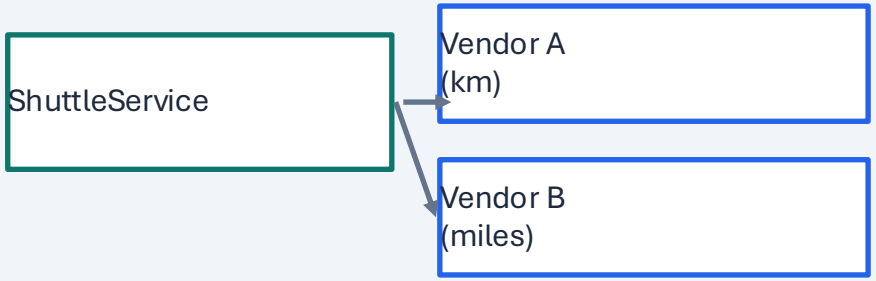
Adapter

Vendor APIs differ (units, shapes, errors) and change over time.

Symptoms

- Two map providers expose different route APIs and units.
- Conversion logic spreads across the codebase.
- Switching vendors breaks many call sites and tests.
- Error formats differ, making failures confusing to handle.

Incompatible interfaces



Unit conversions + error mapping leak into every call site.

Adapter pattern — solution

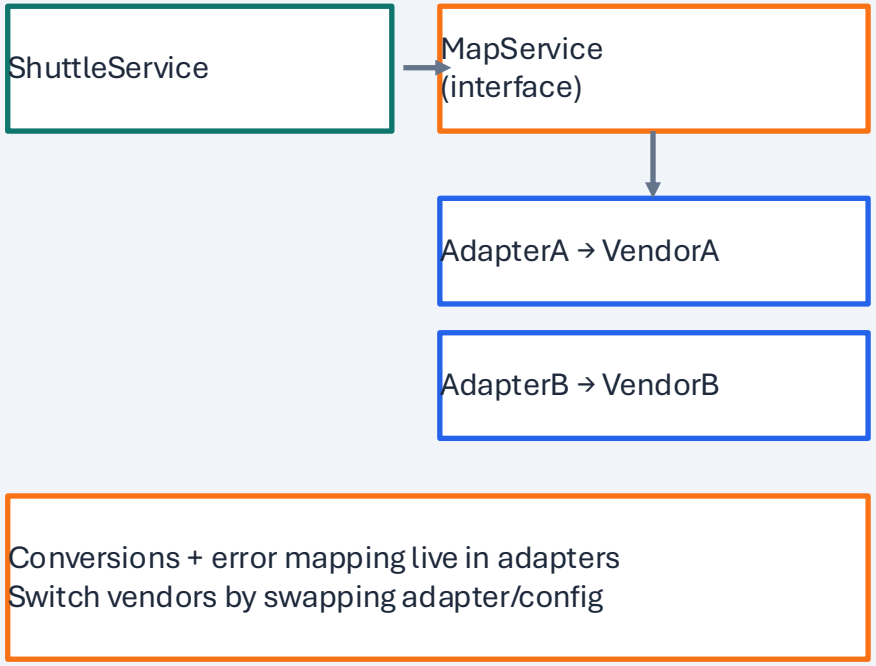
Adapter

Create one stable interface and translate each vendor into it.

How it helps

- Define MapService interface that matches platform needs.
- Adapters translate vendor APIs, units, and errors.
- Switch providers via configuration (not refactor).
- Tests use a fake adapter to avoid external dependencies.

Structure



Adapter pattern — code example

Adapter

One stable interface wraps each vendor; switch providers via config.

```
// 1. Your platform's stable interface
interface MapService {
  getRoute(from: LatLng, to: LatLng): Route;  // always returns km
}

// 2. Adapter for Vendor A (already uses km)
class VendorAAdapter implements MapService {
  getRoute(from, to) {
    const raw = vendorA.directions(from, to);  // vendor-specific call
    return { distanceKm: raw.dist, eta: raw.time };
  }
}

// 3. Adapter for Vendor B (converts miles → km)
class VendorBAdapter implements MapService {
  getRoute(from, to) {
    const raw = vendorB.fetchRoute(from, to);  // different API shape
    return { distanceKm: raw.miles * 1.609, eta: raw.duration }; // convert!
  }
}

// 4. ShuttleService depends only on MapService — never on vendors
class ShuttleService {
  constructor(private maps: MapService) {}  // inject any adapter
  estimateArrival(from, to) { return this.maps.getRoute(from, to).eta; }
}

// Switch vendors by changing one line:
const shuttle = new ShuttleService(new VendorBAdapter());  // swap here
```

Repository pattern — problem

Repository

Business rules leak SQL everywhere; tests get slow and fragile.

Symptoms

- Enrollment logic embeds SQL and query details throughout the codebase.
- Business logic mixes with persistence, making unit tests slow.
- Changing databases (SQLite → Postgres) becomes risky and messy.
- Duplicate queries creep in, causing bugs and inconsistencies.

SQL scattered

```
// inside service layer
db.query('SELECT ... WHERE open=1');
db.query('UPDATE seats SET ...');
// many files, many variants
```

Domain logic + persistence = hard seams

Result: refactors touch too many files.

Repository pattern — solution

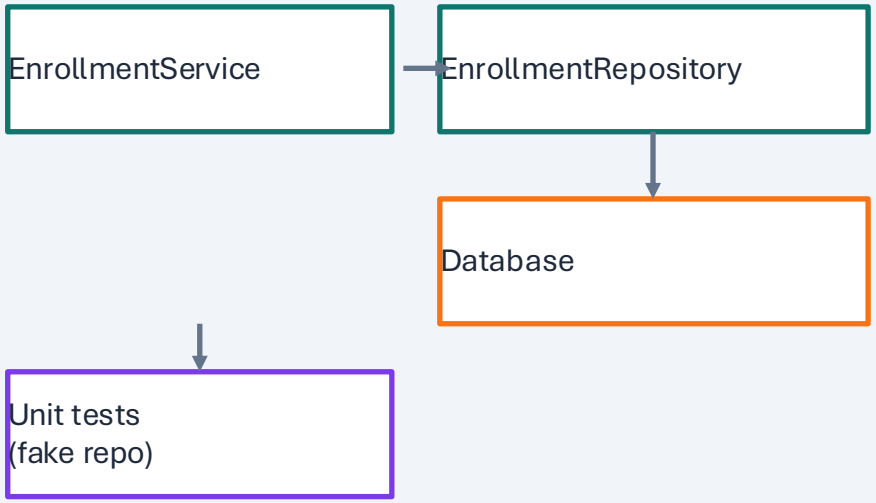
Repository

Expose domain-focused methods; keep SQL and optimization inside.

How it helps

- Repository exposes methods like findOpenSeats().
- Services call a clean API; logic stays readable.
- Mock repositories for fast, reliable unit tests.
- Database changes become localized.

Boundary



Cost: an extra abstraction. Benefit: stability and test seams.

Repository pattern — code example

Repository

Domain-focused methods hide SQL; services stay testable.

```
// 1. Repository interface — domain language, no SQL
interface EnrollmentRepo {
  findOpenSeats(courseId: string): Seat[];
  reserveSeat(courseId: string, studentId: string): void;
}

// 2. Real implementation — SQL lives only here
class PgEnrollmentRepo implements EnrollmentRepo {
  findOpenSeats(courseId) {
    return db.query(`SELECT * FROM seats WHERE course=$1 AND open=true`, [courseId]);
  }
  reserveSeat(courseId, studentId) {
    db.query(`UPDATE seats SET open=false, student=$2 WHERE course=$1`, [courseId, studentId]);
  }
}

// 3. Service uses repo — no SQL, easy to test
class EnrollmentService {
  constructor(private repo: EnrollmentRepo) {}

  enroll(courseId: string, studentId: string) {
    const seats = this.repo.findOpenSeats(courseId);
    if (seats.length === 0) throw new Error("No seats");
    this.repo.reserveSeat(courseId, studentId);  // clean domain call
  }
}

// 4. In tests: use a fake repo — no database needed
const fakeRepo: EnrollmentRepo = {
  findOpenSeats: () => [{ id: "s1", open: true }],
  reserveSeat: () => {},
};
const svc = new EnrollmentService(fakeRepo);  // instant, no DB
```

Command pattern — problem

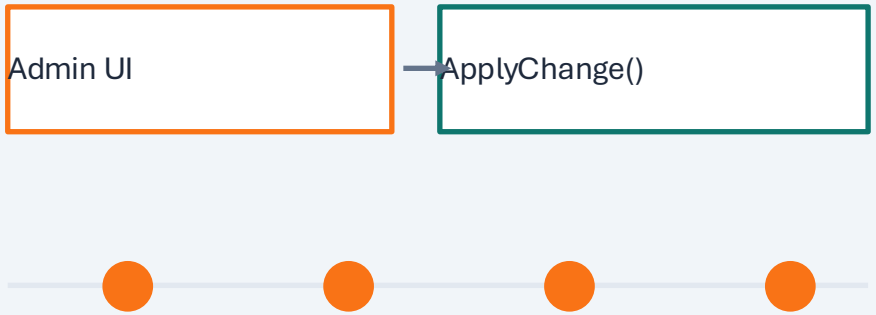
Command

Admins need audit, but direct calls lose intent.

Symptoms

- Coupling the invoker of a request to a particular request should be avoided.
- Auditing means that every single direct call should also implement the audit
- Audit changes mean whole code base should be refactored
- Hard-wired requests should be avoided.
- It should be possible to configure an object (that invokes a request) with a request.
- Direct calls apply changes without a record of intent.

No undo trail



Change history? (missing)

Command pattern — solution

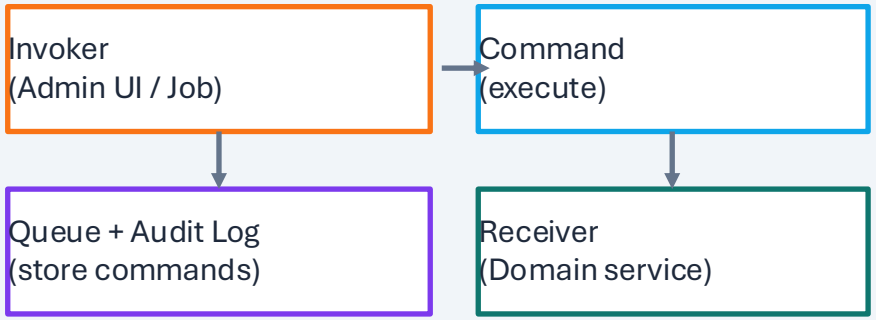
Command

Encapsulate actions as objects with execute() and undo().

How it helps

- Passing the request as input allows for audit and
- Each action is a Command: execute + undo.
- Commands can be queued, logged, and retried safely.
- Audit log stores commands (intent), not just final state.
- Small units make testing easier.

Structure



Tradeoff: more objects. Benefit: traceability + reversibility.

Command pattern — code example

Command

Encapsulate actions as objects with execute() and undo().

```
// 1. Command interface
interface Command {
  execute(): void;
  undo(): void;
  description: string;  // for the audit log
}

// 2. Concrete command — captures intent + supports undo
class DropStudentCmd implements Command {
  description = "Drop student from course";
  constructor(private courseId: string, private studentId: string) {}

  execute() { enrollmentService.drop(this.courseId, this.studentId); }
  undo()    { enrollmentService.enroll(this.courseId, this.studentId); } // reverse!
}

// 3. Invoker stores commands for audit + undo
class AdminInvoker {
  private history: Command[] = [];

  run(cmd: Command) {
    cmd.execute();
    this.history.push(cmd);  // audit trail
    console.log(`[AUDIT]`, cmd.description);
  }
  undoLast() { this.history.pop()?.undo(); }  // undo!
}

// 4. Usage
const admin = new AdminInvoker();
admin.run(new DropStudentCmd("CS101", "stu42"));  // logged + reversible
admin.undoLast();  // mistake? just undo
```

State pattern — problem

State

Workflows evolve over time; scattered transitions create bugs.

Symptoms

- Ride status: requested → matched → en route → completed.
- Conditionals for transitions spread across the codebase.
- Invalid transitions slip in (e.g., complete without match).
- Adding a new state means touching many if/else blocks.

Scattered transitions

```
if (ride.status === 'requested') {
  // ...
} else if (ride.status === 'matched') {
  // ...
} else if (...) {
  // ...
}
```

Same logic duplicated in many places

Hard to ensure “only valid moves”.

State pattern — solution

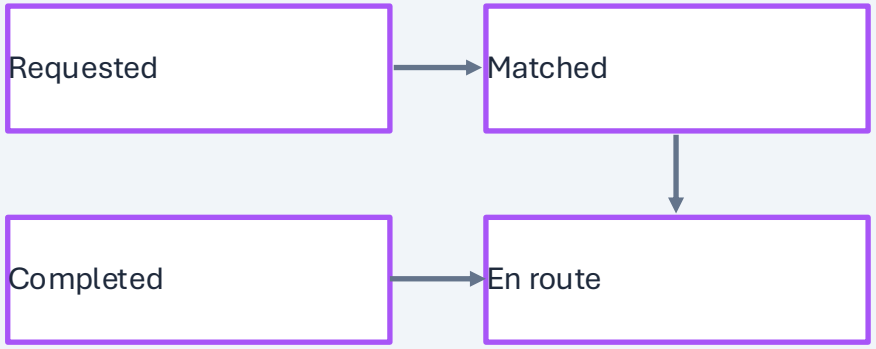
State

Put behavior + transitions inside explicit state objects.

How it helps

- Each state object defines allowed transitions.
- Invalid transitions are blocked by design.
- New states = add a class (localized change).
- Tests focus on transitions (higher confidence).

State machine (example)



Only allowed transitions exist.

State pattern — code example

State

Each state object defines allowed transitions; invalid moves are blocked.

```
// 1. State interface
interface RideState {
  match(driverId: string): RideState;
  startRide(): RideState;
  complete(): RideState;
}

// 2. Concrete states — each defines valid transitions only
class Requested implements RideState {
  match(driverId)  { return new Matched(driverId); }  // valid
  startRide()       { throw new Error("Not matched yet"); }  // blocked!
  complete()        { throw new Error("Not started yet"); }  // blocked!
}
class Matched implements RideState {
  constructor(private driverId: string) {}
  match()         { throw new Error("Already matched"); }
  startRide()     { return new EnRoute(this.driverId); }  // valid
  complete()      { throw new Error("Still en route"); }
}

// 3. Ride context delegates to current state
class Ride {
  private state: RideState = new Requested();
  match(id: string) { this.state = this.state.match(id); }
  start()             { this.state = this.state.startRide(); }
}

// ride.start() before ride.match() → throws "Not matched yet"
```

Strategy vs State — how to choose

Input-based variability vs lifecycle-based variability.

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Decision cue

Ask: “Why does behavior change?”

- Strategy: choose behavior based on input/context.
- State: behavior changes over time as the object evolves.
- Avoid mixing unless you can clearly name both forces.

Quick comparison

Pattern	Best fit
Strategy	Pricing/eligibility rules, pluggable algorithms
State	Workflows, lifecycles, “valid transitions”

Tip: if you draw a state diagram, you probably want State.

Combining patterns in one feature

Patterns stack to reduce risk across logic, data, and notifications.

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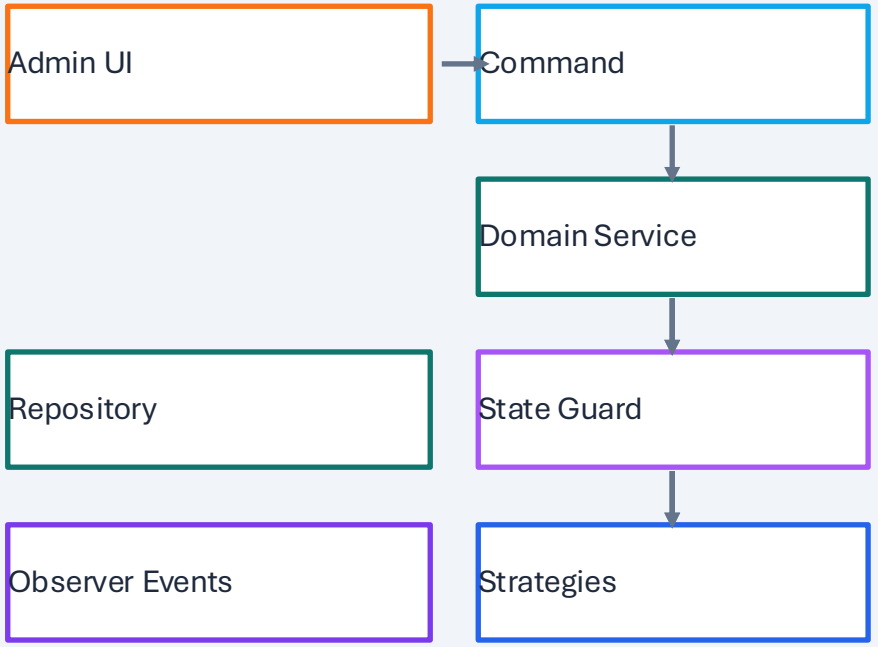
Example: enrollment change

- Command captures admin intent and enables undo + audit.
- Repository isolates persistence and makes side effects visible.
- State guards valid phases of an enrollment.
- Observer notifies students and analytics after commit.
- Strategy selects pricing/eligibility rules by context.

Architecture sketch



One feature, multiple seams (each testable on its own).

Adopt patterns safely

Incremental refactors + a simple review checklist.

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Refactoring paths (incremental)

1) If/Else → Strategy

- Write tests for current outcomes
- Extract one rule into a strategy
- Replace branches step-by-step

2) Direct calls → Observer

- Add an event object
- Emit event without changing behavior
- Move one subscriber at a time + add correlation IDs

3) Inline SQL → Repository

- Inventory queries by domain concept
- Move SQL into repository methods
- Test repos + services separately

Code review checklist + takeaway

- Does this solve a real recurring problem here?
- Is it the smallest pattern that fits?
- Are responsibilities clearer and more testable?
- Are tradeoffs documented (e.g., ADR)?

Lessons learned

- Use fewer patterns well; avoid pattern “style theater”.
- Always ask what cost you are paying for flexibility.
- Patterns are easiest to adopt via safe, incremental refactors.

Document decisions with short ADRs (problem → decision → consequences).